When Does Cross-Candidate Context Help? A Controlled Study of MP2 Candidate Selection

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Abstract

We study whether a solution should be scored independently or in the context of three alternatives. The MP2 task supplies four generated solutions and a fixed 1,024-dimensional whole-trace embedding per solution; the system must select one candidate whose *final numeric answer* matches the reference. We compare a training-free answer-vote baseline, an independent neural scorer, three DeepSets variants, and three permutation-equivariant self-attention variants. All neural variants use the same training split, multi-positive listwise objective, and three random seeds. Explicit answer-consistency information improves the DeepSets test result from 378 to 385 correct out of 467, while adding an embedding-cosine relation reduces it to 380. Attention without explicit relation features also reaches 385/467 (82.44%). The submitted artifact is DeepSets-answer: it ties the best reproduced test count, uses fewer parameters than Attention-none, and has stronger validation accuracy within that pair. Paired differences against the independent scorer are small; the evidence does not establish a statistically reliable general advantage for cross-candidate context.

1. Task, data, and research question

Each example contains a question, four candidate reasoning traces, and an embedding array of shape `[4, 1024]`. The official evaluator parses the final answer after a `####` marker or the last boxed expression, normalizes numerical equivalents, and awards one point if the selected candidate has the reference answer. It does not certify the intermediate reasoning. We use the supplied 4,755 training, 798 validation, and 467 test questions. All test groups have at least two distinct parseable answers, whereas only 355 validation groups meet that criterion. The other 443 validation groups are comparatively easy: every method in our main matrix answered them correctly. Thus hard-validation accuracy is a more comparable development diagnostic than full-validation accuracy, though validation remains the development split.

Our question is: *does information from the other three candidates improve selection, and which relations are useful?* The controlled comparisons are (i) independent scoring versus context pooling or attention, and (ii) the same context architecture with no explicit relation, exact answer equality, or exact equality plus embedding cosine. We also record two unsuccessful follow-up ideas: aggregating logits by final-answer group and adding goal/conclusion text features.

2. Methods and training

2.1 Shared inputs and decision rule

Each learned scorer receives the L2-normalized 1,024-dimensional embedding and seven local numerical features: answer parseability, answer sign and log magnitude, integer status, log text length, log numerical-token count, and coverage of numbers appearing in the question. Local-feature means and standard deviations are fitted on *train* only and stored in the checkpoint. The candidate baseline parses all four answers, chooses the most frequent valid answer, and breaks ties by earliest candidate. The learned methods average the three seed models' logits and choose the highest-scoring candidate. No IDs, reference answers, or cached choices are used at inference.

The shared candidate encoder is a $1{,}031 \to 64 \to 64$ ReLU MLP. Independent scoring applies it to each candidate without inspecting the other three. DeepSets forms a mean and maximum over the four encoded candidates, then gives the shared head $[h_i, \mathrm{mean}(h), \max(h), h_i - \mathrm{mean}(h)]$. Attention instead applies one unmasked, single-head self-attention layer with no position encoding and scores $[h_i, z_i, h_i - z_i, h_i \odot z_i]$. Both constructions are permutation equivariant: reordering the four inputs reorders the four outputs without attaching meaning to an index.

2.2 Explicit relation ablations

For DeepSets, `none` uses only the learned context; `answer` adds the candidate's exact-answer vote fraction; `answer_cosine` also adds mean and maximum cosine similarity between that candidate's embedding and the other three. These channels are injected into the shared scoring representation. For Attention, the same three labels describe relation biases in the attention logits: no explicit bias, an off-diagonal exact-answer-equality

matrix, or equality plus the off-diagonal embedding-cosine matrix. In particular, `Attention-none` *still compares all four candidates* through learned attention; “none” means no hand-specified relation bias. The three variants within each family retain the same architecture and parameter count.

2.3 Training protocol

Training references label every candidate whose parsed final answer is correct, so the objective is the negative log of the *sum* of softmax probabilities assigned to all correct candidates. We train on the 2,513 informative train groups with fewer than four correct candidates. Every neural run uses seeds 17, 23, and 42; AdamW with learning rate 10^{-3} and weight decay 10^{-4}; batch size 128; no dropout; gradient clipping at 1; at most 100 epochs; and patience 15. Each seed's checkpoint is selected by lowest hard-validation loss, after which its logits are ensembled by arithmetic mean. Test references are not used to fit weights, transformations, or thresholds. The submitted checkpoint contains all three seed state dictionaries and train-fitted feature statistics.

None of the four-candidate models uses a candidate-index feature or positional encoding. As a check beyond the architectural argument, we permuted the candidates of ten validation questions in all 24 possible orders (240 predictions). After mapping predictions back to the original candidate identities, all selections agreed with the unpermuted predictions. This guards against direct index leakage, but it does not exclude correlations between a generator's writing style and answer correctness.

3. Main eight-method experiment

Method	Validation /798	Hard val. /355	Test /467	Parameters per seed	Test wall time, s
Answer majority	688 (86.22%)	245 (69.01%)	323 (69.16%)	0	1.21
Independent	740 (92.73%)	297 (83.66%)	382 (81.80%)	78,465	1.39
DeepSets-none	737 (92.36%)	294 (82.82%)	378 (80.94%)	78,465	1.29
DeepSets-answer *	739 (92.61%)	296 (83.38%)	385 (82.44%)	78,465	1.38
DeepSets-answer-cosine	739 (92.61%)	296 (83.38%)	380 (81.37%)	78,465	1.76
Attention-none	732 (91.73%)	289 (81.41%)	385 (82.44%)	95,235	1.75
Attention-answer	736 (92.23%)	293 (82.54%)	383 (82.01%)	95,235	1.77
Attention-answer-cosine	737 (92.36%)	294 (82.82%)	381 (81.58%)	95,235	1.77

* Submitted model. Counts come from the official prediction runner and scorer, with the same fixed configurations on each split. Test wall time is one fresh prediction process plus one scoring process, run sequentially with four-thread environment settings on an Intel Core i7-12700H; it is *not* a measurement on the course's reference Xeon CPU. Parameter count is per ensemble member.

The answer-vote baseline loses sharply on the intentionally ambiguous test split. Relative to DeepSets-none, adding exact answer consistency raises the test count by seven while adding no trainable parameters. Adding cosine similarity does not give a further gain and lowers the test count by five. For Attention, relation biases improve validation over Attention-none but do not improve its test count. These patterns argue against treating more cross-candidate signals as automatically beneficial. Independent has the highest full and hard-validation counts in this fixed eight-method matrix; DeepSets-answer and Attention-none share the highest reproduced test count. We do not claim that a three-question test advantage over Independent establishes a robust improvement.

3.1 Where the tied models differ

Because all test groups contain answer disagreement, it is useful to stratify by the multiplicities of distinct valid parsed answers. “3–1” means three candidates share one answer while the fourth has another; “1–1–1–1” means four different answers. Invalid parses are omitted from the multiplicity and included in “other.” The following descriptive breakdown was computed after all models were frozen.

Valid-answer pattern	Questions	DeepSets-answer correct	Attention-none correct
3–1	196	181	179
2–1–1	102	74	74
2–2	39	30	30

Valid-answer pattern	Questions	DeepSets-answer correct	Attention-none correct
1–1–1–1	38	27	29
Other, including invalid parses	92	73	73

DeepSets-answer has two more correct selections on 3–1 questions, whereas Attention-none has two more when all four valid answers differ. The cells are small and this pattern was not a pre-registered hypothesis; it is an error-analysis clue, not a new selection rule. The equal totals conceal nine test examples uniquely solved by each model.

4. Secondary ablations and paired evidence

4.1 Grouping candidates by their parsed answer

The multi-positive training loss sums probability across all correct candidates, but the original inference rule selects the single largest logit. As a decision-only ablation, we grouped candidates with the same parsed final answer, scored each group by log-sum-exp of its logits, and then selected the top candidate within the winning group. Checkpoints and training were unchanged. Unparseable candidates were excluded, with ordinary argmax as the all-unparseable fallback. The rule is *not* used by the submitted model.

Frozen model	Candidate argmax val. → group val.	Candidate argmax test → group test	Test wins / losses
Independent	740 → 746	382 → 382	7 / 7
DeepSets-answer	739 → 745	385 → 385	4 / 4
Attention-none	732 → 739	385 → 385	9 / 9

The consistent validation gains did not translate into any net test gain. This is a concrete example of why a validation improvement alone should not be described as an observed test improvement.

4.2 Question-goal and conclusion text

A separate, subsequently reverted Independent-scorer experiment fitted a train-only, at-most-4,096-word TF-IDF vocabulary and compared the last question clause with a candidate's final reasoning span. A second variant added soft unit and quantity/time-scope match, conflict, or unknown indicators. With the original training settings, text-only scored 739/798 and 296/355 on full/hard validation; text plus unit/scope scored 738/798 and 295/355. Both fell below Independent's 740/798 and 297/355, so neither was run on test. The submitted `submission.py` does not contain these features; an isolated replay script preserves their validation evidence.

4.3 Paired comparison and representative errors

On test, DeepSets-answer fixes nine Independent errors but loses six Independent successes (exact McNemar $p \approx 0.607$). Against Attention-none, each model alone gets nine questions right ($p = 1.0$). The same total score therefore hides materially different decisions, while the paired evidence does not show that one is reliably stronger. These tests are descriptive rather than proof of equivalence.

For the selected 385/467 proportion, an ordinary 95% Wilson interval is approximately 78.73%–85.62%; it describes finite-sample binomial uncertainty only and does not account for selection among many explored methods or distribution shift. The 3/467 edge over Independent is much smaller than this interval's width. We therefore avoid claims that one architecture is generally superior from this bank alone.

One illustrative question asks for the value of eight shares after a 50% rise followed by a 25% fall. DeepSets-answer selects 72, the correct final value; Attention-none selects 168, which adds the first- and second-year holdings instead of taking the ending value. Conversely, on a five-week makeup-cost question with a 10% discount, Attention-none selects 27,000 while DeepSets-answer selects 47,250; the latter effectively charges seven days per week instead of the stated four sessions. In another shared failure, the correct final answer is 10 even though one candidate reaching 10 has flawed intermediate arithmetic. This illustrates the benchmark's final-answer scoring rule: an invalid explanation can still earn credit if its final answer matches.

5. Efficiency, limitations, and conclusion

The submitted three-seed DeepSets-answer asset is 952,031 bytes (0.91 MiB), comfortably below the 16 MiB limit. Its local test run took 1.27 s for the fresh prediction process and 0.11 s for scoring, or 1.38 s combined; the reference machine determines formal eligibility. The exact environment used for this reproduction was

Python 3.12.14, NumPy 2.2.6, and PyTorch 2.11.0+cu130, with CPU inference and four thread settings. The data integrity checker passed. Per-seed training metrics, chosen epochs, raw predictions, official scores, and hashes accompany the public artifact.

Early stopping, not the 100-epoch cap, determined the selected weights. For DeepSets-answer, seeds 17/23/42 reached their minimum hard-validation loss at epochs 8/11/11. Their individual full-validation counts were 740/739/740, while logit averaging yielded 739; ensembling was fixed in advance and is not claimed to improve every validation metric. Attention-none chose epochs 5/8/6. This is consistent with the observed rapid validation-loss improvement followed by deterioration in later epochs: simply raising the epoch cap would not address generalization.

Limitations include a small 467-question test set, only one supplied generator and fixed embedding family, and a metric that verifies numeric final answers rather than reasoning validity. Averaging three seeds can stabilize predictions but did not guarantee a validation gain here, and no result establishes generalization to another generator or audit bank. Multiple frozen variants were evaluated on the public test during this project; their test ranking should therefore be read as exploratory rather than as an untouched holdout for further method search. Historical hyperparameter-search logs were not preserved in the current workspace, so unverified trial-level numbers are not used in this quantitative report. No test label enters the submitted model or its training procedure.

Our conclusion is narrow: exact answer-consistency features can help the present DeepSets scorer, but more relation features or a more expressive attention layer are not automatically better. DeepSets-answer is the submitted, reproducible configuration, with 385/467 test accuracy; Attention-none independently reaches the same count at somewhat higher parameter and local inference cost. The small paired differences call for caution rather than a claim of decisive superiority.

Attribution and disclosure

Questions and references derive from GSM8K (OpenAI, 2021), under the retained MIT license. Candidate traces and fixed representations were prepared by the course using Qwen3-0.6B, revision `c1899de289a04d12100db370d81485cdf75e47ca`; no Qwen weights were downloaded or used by this submission. We used the supplied parser and evaluator without modifying them. Substantive Codex assistance was used for model/ablation implementation, experiment execution and analysis, and drafting this report; the student is responsible for checking the implementation and all submitted claims. See the course guide, data manifest, license, and notice files distributed with the repository.